Jinhun Heo

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Education

The University of Texas at Austin Expected May 2029
BS in Mechanical Engineering; Minor in Robotics GPA: 3.79/4.00

Skills

Design & Manufacturing: OnShape, SolidWorks, Fusion 360, FDM 3D printing, Machining
Programming: Python, C++, Arduino, Linux, ROS, Git

Work Experience

Undergraduate Research Assistant, Robotic Vision Lab – The University of Texas at Arlington June 2026 – Aug 2026

- Engineered mounting solutions with sliding alignment rails and integrated Velcro-strap slots for sensors and communication hardware, minimizing fasteners to achieve a 67% reduction in hardware assembly time
- Designed custom snap-fit clips to securely attach electronics to metal rails on the Husky UGV for high-vibration tasks while enabling easy detachment for rapid troubleshooting, reducing unplanned downtime
- Modeled a microcontroller enclosure with side-clamping snap-fits, creating overhead protection for GPIO pin headers and PCB signal traces while leaving enough openings for easy pin reconfiguration and monitoring of status LEDs
- Reduced hardware failure risks by integrating custom fan grills to block debris entering internal UGV wiring, along with custom cable-management parts to organize and protect exposed stereo camera cables during transport
- Constructed an indoor sensor testing/calibration rig enabling rapid laboratory verification without full UGV assembly

Projects

Reaction Wheel, Freshman Research Initiative (Space & Robotics Stream) Jan 2026 – May 2026

- Implemented and tuned a closed-loop PID control system bringing a hand-spun platform to a stop within 0.5s
- Maintained 89% continuous tracking on colored targets experiencing violent shaking by developing a dual-loop PID control architecture integrated with real-time OpenCV tracking
- Generated a web-based monitoring interface using Flask to stream live camera footage, creating graphical overlays such as a crosshair and a bounding box to visually verify real-time target tracking performance
- Designed a weight-optimized motor housing, reducing battery consumption by 11% and tipping by 29%
- Achieved bidirectional motor control and PWM duty cycle modulation by introducing an H-bridge
- Mitigated thermal risks and optimized power efficiency by sizing wire gauges and calculating peak current loads

Human Re-Identification, Learning Agents Research Group Dec 2025 – May 2026

- Engineered a real-time human re-identification system on the Booster T1, integrating two CNN deep learning models and a tracking algorithm on ROS2 for the RoboCup@home competition
- Developed ID maintaining logic utilizing spatial tracking and ID lifespan, resulting in 80% reduction in ID switching
- Introduced face recognition to augment profiles, enabling ID recovery in lighting variations and clothing changes
- Optimized performance by leveraging CUDA for PyTorch-based algorithms achieving stable 22 fps processing
- Created a real-time monitoring interface, broadcasting bounding box data for validation and debugging

Baja SAE Pedalbox, Longhorn Baja Racing Sept 2025 – Apr 2026

- Spearheaded the design of the throttle pedal assembly using SolidWorks, integrating a custom throttle cable linkage to navigate a rear-engine architecture while preventing mechanical interference with mid-chassis components
- Optimized pedal arm length to 9in, a 25% reduction from industry standards, meeting competition regulations
- Resolved lack of pedal return by introducing spring mechanism, ensuring reliable return and driver feedback